POLYNEXUS HOSPITAL MANAGEMENT SYSTEM

ABDM Sandbox Readiness

Where the ABDM integration stands and the path to NHA certification

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THE JOURNEY AT A GLANCE

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1. Where we stand	What is built, what isn't, in one table
2. The certification path	The steps NHA expects, in order
3. Work still to build	Engineering items, with size and order
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Before you start

ABDM (Ayushman Bharat Digital Mission) integrations are certified by the National Health Authority (NHA) through its sandbox before production credentials are issued. **Confirm the current steps on the ABDM sandbox portal before starting**: NHA updates the milestones, API versions and audit requirements from time to time.

1 Where we stand

ABDM certification is usually done in three milestones:

Milestone	Means	Status in our product
M1: ABHA	Create or verify a patient's ABHA number and link it to the hospital's patient record	**Partly built.** Verification by Aadhaar OTP, mobile OTP or existing ABHA, and linking to the UHID, are built (`AbhaLink`). Creating a new ABHA from inside our app is not built. Field names are best-effort and must be checked against live sandbox responses.
M2: Health Information Provider (HIP)	The hospital shares records out: patients find and link their records (care contexts), approve consent, and the hospital pushes encrypted FHIR records	**Not built.** FHIR document bundles are built (prescription, lab report, imaging report, discharge summary, OP consultation; ABHA included). Missing: care-context discovery and linking callbacks, consent notification handling, and the encrypted data push (ECDH key exchange).
M3: Health Information User (HIU)	The hospital asks for a patient's records held elsewhere, with their consent, and receives them	**Partly built.** Consent requests and status checks are built (`ConsentRequest`). Receiving the records (an asynchronous, encrypted push to our callback) is not built.
NHCX (insurance claims)	Claims through the National Health Claims Exchange	**Partly built.** `NHCXTransaction` and its API exist; the live gateway is a stub until onboarding.

Today every ABDM call goes to a **stub** gateway unless `ABDM_GATEWAY=real` and sandbox credentials are set, so nothing has yet been run against a live ABDM response.

NABH elements this affects: AAC.1.e (ABHA capture), AAC.3.n / AAC.4.m / AAC.6.f / COP.1.o (records linked to ABHA: bundles built and tested), COP.1.p (access to past records through ABHA: partial), IMS.1.b–d (extended FHIR profiles and NHCX: partial or gap).

2 The certification path

1. **Register** on the ABDM sandbox as the integrator, and get sandbox client credentials.
2. **Register the facility** in the Health Facility Registry (HFR) and the doctors in the Healthcare Professionals Registry (HPR), as needed for HIP/HIU IDs.
3. **Build and self-test** each milestone against the sandbox (Part 5).
4. **Functional testing** by NHA's team in the sandbox: they run the scenarios and check the screens and responses.
5. **Security audit:** ABDM asks for a web-application security audit report from a CERT-In-empanelled auditor before production. Check the current requirement and plan the audit early: findings take time to fix.
6. **Production** credentials and go-live, with the facility's production IDs.

3 Work still to build

In the order to do it:

#	Item	Why	Size
1	Run M1 against the sandbox and fix field names	Our ABHA calls are built from the published API collection, not live responses	Small
2	ABHA creation (not only verification)	Patients without an ABHA should be able to get one at registration	Medium
3	Gateway callback endpoints (signed requests from ABDM)	ABDM's flows are push-based: it calls our `on-init`, `on-status`, `on-notify` and similar endpoints	Medium
4	HIP care-context discovery and linking	M2: patients link their visits at our hospital to their ABHA	Medium
5	Encrypted data push (ECDH/X25519 with the published encryption spec)	M2 and M3: health records travel encrypted end to end	Large
6	HIU data receipt and display	M3: show records fetched from other providers	Medium
7	Extended and advanced FHIR profiles (immunisation, wellness, health documents)	NABH IMS.1.b/c	Medium
8	NHCX live gateway	Insurance claims (FPM.4.i, IMS.1.d) after NHCX onboarding	Medium

4 What the hospital or company must do

These are not software features, and certification can't proceed without them:

- The sandbox account and credentials, kept secret (they go into the server's `.env`, never into code).
- HFR registration of each hospital, and HPR IDs for the doctors who will sign records.
- A public HTTPS address for the callback endpoints (ABDM must be able to reach them).
- The security audit by a CERT-In-empanelled auditor, and fixing its findings.
- Patient-facing consent wording for ABDM sharing: the consent register's **ABDM sharing** notice (Security & Governance → Consent notices).
- A named person for NHA's communication, and for incidents.

5 How we will test in the sandbox

1. Set `ABDM_GATEWAY=real`, `ABDM_BASE_URL`, `ABDM_CLIENT_ID`, `ABDM_CLIENT_SECRET` and `ABDM_HIP_ID` on a test server (never on a hospital's production server).
2. **M1:** verify a sandbox test ABHA by mobile OTP and by Aadhaar OTP; link it to a test patient; check the ABHA appears in the patient record and in the FHIR bundles.
3. **M3 consent:** raise a consent request for a sandbox ABHA, approve it in the sandbox PHR app, check our status changes to granted.
4. **M2:** discover and link care contexts from the sandbox PHR app; approve a consent; push a prescription, lab report and discharge summary; confirm the sandbox decrypts and validates them.
5. **M3 data:** receive another provider's records and show them in the patient's record.
6. Keep the request IDs and screenshots of each step: NHA's functional testing asks for them.

6 Where the code is

Area	Location
Gateway client (stub and real)	`Backend/apps/abdm/gateway.py` : its docstring lists exactly what is grounded and what is not
ABHA link, consent requests, record fetch, NHCX	`Backend/apps/abdm/models.py` , `views.py`
FHIR document bundles	`Backend/apps/integrations/abdm_fhir.py`
Tests	`Backend/apps/abdm/tests.py` , `Backend/apps/integrations/test_abdm_fhir.py` , `Backend/apps/quality/test_nabh_core.py` (imaging bundle carries the ABHA)
Settings	`ABDM_GATEWAY` , `ABDM_BASE_URL` , `ABDM_CLIENT_ID` , `ABDM_CLIENT_SECRET` , `ABDM_HIP_ID` , `ABDM_TIMEOUT_SECONDS`